

Backend sprints

Sprint 1: Foundation & Database Setup

Goal: To lay out the foundation of the project and configure the database.

- Create a C# .NET backend project and configure Entity Framework Core to work with it.
- Connect backend project to Neon.tech PostgreSQL cloud database.
- Configure entity models for 8 core tables: Users, Subscriptions, Transactions, Visits, Exercises, Routines, RoutineLists, and Equipment.
- Execute database migrations to build the corresponding tables in the DB.

Sprint 2: Authentication & Security

Goal: To make the backend secure and provide the functionality to register and authenticate the user.

- Configure backend with .NET Identity Framework for hashed password storage.
- Implement backend API services for User Registration and Login via Email/Password.
- Implement RBAC to establish a security boundary between the roles of 'Admin' and 'Member'.

Sprint 3: Core Administration Operations

Goal: To make backend services which allow the gym owner to administrate facilities.

- Implement CRUD endpoints for Member Management allowing the Admin to manipulate the user's properties such as status, name, email, etc.
- Implement CRUD endpoints for Equipment Management allowing the Admin to record a new piece of equipment or schedule it for maintenance.
- Restrict access to Member Management and Equipment Management endpoints to users with the 'Admin' role.

Sprint 4: Subscriptions & E-Commerce

Goal: To process payments for subscriptions.

- Implement a Checkout endpoint to receive payment requests from the frontend (Day Pass/ Monthly Pass).
- Implement generation of the Transaction when payment is successfully processed.
- Implement database updates of **SubscriptionEndDate** and **PaymentStatus** properties after a subscription purchase.

Sprint 5: Geolocation & Attendance Tracking

Goal: To implement logic that allows verifying the user's location and creating a new Visit.

- Create a Check-In endpoint that will accept the user's QR code information from the frontend along with GPS coordinates.
- Implement a geofencing algorithm that will confirm the GPS coordinates are within 50 meters from the gym's actual location.
- If verified successfully, generate a new Visit record in the database. Otherwise – return an error message.

Sprint 6: Automated Workout Engine

Goal: To develop an automated workout generator which creates daily workouts.

- Implement an endpoint to accept the preferred training split from the user as an input parameter (for example, Push/Pull/Legs, Full Body, etc.).
- Implement backend query logic to filter out exercises from the database which correspond to the training split selected.
- Create a daily workout plan using the filtered-out exercises and save the routine in the **Routines** and **RoutineLists** tables.

Sprint 7: Business Analytics Engine

Goal: To gather relevant business intelligence about customers and transactions.

- Create a Reporting Engine using LINQ queries to aggregate backend data.
- Write a query to calculate total monthly revenue from the **Transactions** table.
- Write a query to analyze **Visit** timestamps and generate peak usage hour histograms.
- Create an endpoint to send aggregated business intelligence safely to the Admin dashboard.

Frontend sprints

FRONTEND SPRINT PLAN

[MOBILE LAYOUT]

Week 1: Setup Basic Pages (landing page, registration page, sign-in page)

Step 1: Create the main project folder and set up code rules for the team.

Step 2: Layout the brand colors like black backgrounds and gold/yellow accents to the system settings. (reference the palette from the wireframe)

Step 3: Landing Page

- Set up the full-screen canvas view by rendering a solid matte black background layer detailed with subtle, dark geometric honeycomb hex-grid element at the opposing corners
- Center the primary branding component (Kabakal Gym's Logo) directly in the upper middle section of the screen.
- Add the central typography element by positioning the yellow subtitle text statement, "Sign in to unlock your daily programming", right above a decorative thin gold horizontal separating line template.
- Create the primary navigation touch targets by stacking two pill-shaped buttons at the bottom: an upper border-outlined button reading "Sign In" and an identical lower button reading "Create Account".

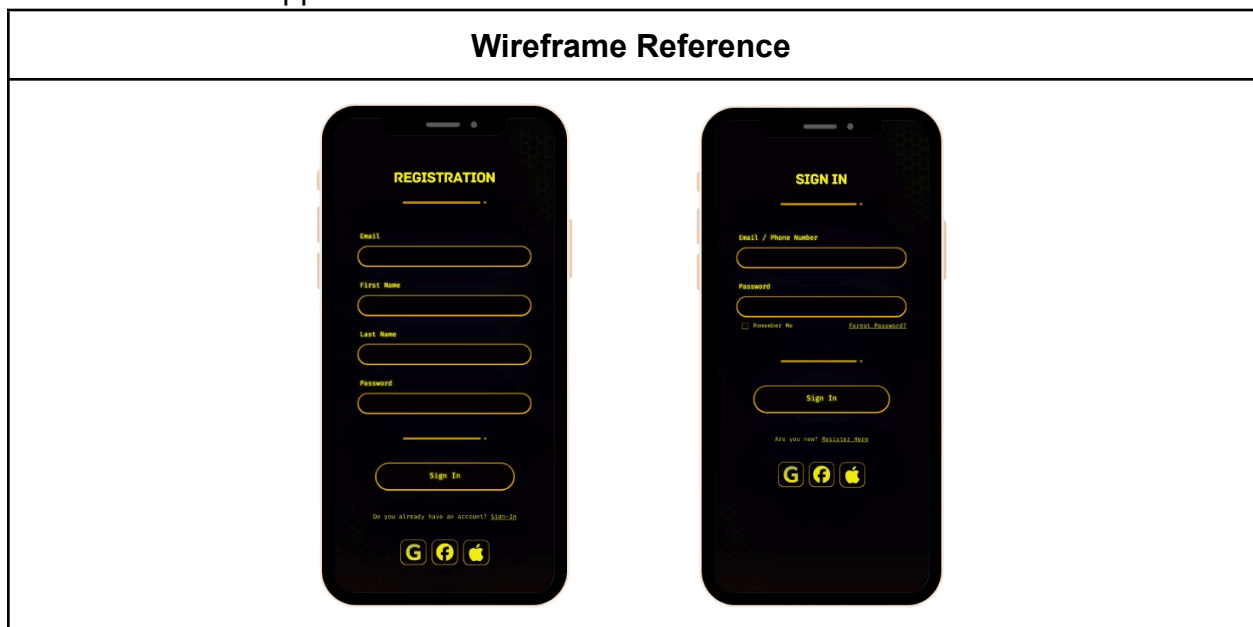


Step 4: Set up the links between pages so users can click through the app.

Note: Do this along the progress to connect all link pages

Step 5: Registration Page (Create Account), Sign In Page (Sign-in)

- Set up the full-screen view layout using a solid black canvas background accented with dark geometric honeycomb hex-grid watermarks positioned in the opposing corners.
- Create the top structural section by centering bold, yellow text headers for either "REGISTRATION" or "SIGN IN" directly over a decorative gold dividing underline component.
- Construct the form input stack by positioning gold-bordered, pill-shaped input boxes under yellow label texts for Email, First Name, Last Name, and Password on the signup screen, or Email/Phone Number and Password on the login screen.
- Incorporate form accessory elements right below the inputs, adding a "Remember Me" checkbox and a "Forgot Password?" text link on the login layout, followed by a primary gold-bordered "Sign In" action button.
- Assemble the bottom navigation and alternative access area by placing a simple redirect text link ("Register Here" or "Sign-In") directly above a row of three rounded square OAuth social login buttons for Google, Facebook, and Apple credentials.



Step 6: Connect the signup and login forms to the backend server so user accounts can be created and checked.

Week 2: Workout Generator Page and the Components of Navigation bar

Step 7: WorkOut Generator Page

- Set up the workout generator page where users answer fitness choices after logging in.
- Add goal choices like building muscle, gaining strength, or losing weight.
- Add choices for users to pick what target split routine they want to do (e.g., Arm day, Leg Day, Full Body, etc.)
- Add simple options to select workout times like 15-30 minutes or 60 minutes.
- Create an easy slider tool on the tab to choose beginner or advanced experience levels.
- Add the generate button to submit the answers and load the main app layout (Home Page).

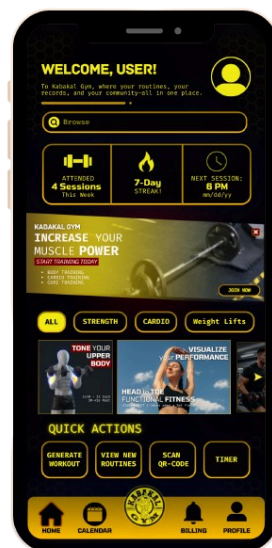
Wireframe Reference

The wireframe shows a smartphone screen with a dark background and yellow text and buttons. The title 'WORKOUT GENERATOR' is at the top. Below it is a section 'A. SELECT FITNESS GOALS' with four buttons: 'Hypertrophy (Muscle Growth)', 'Pure Strength (Weight Lifting)', 'Endurance and Fat Loss', and 'Other'. The next section is 'B. TARGET SPLIT ROUTINE' with four buttons: 'Leg Day', 'Arm Day', 'Full Body', and 'Other'. The final section is 'C. TRAINING DETAILS' with a dropdown menu for 'AVAILABLE SESSION TIME' set to '15-30 Minutes' and a row of five radio buttons for 'EXPERIENCE LEVEL' labeled 'BEGINNER', followed by four unlabeled buttons, and 'PRO'. The 'BEGINNER' radio button is selected. At the bottom is a large 'GENERATE' button.

Step 8: Home Page

- Build a top banner area with the welcoming user header text, profile icon avatar, and text search input bar, alongside a solid gold or yellow-gradient bottom navigation bar housing home, calendar, billing, and profile view items.
- Construct the horizontal metric dashboard container grouping three statistical data tracking components: an attended workout sessions accumulator card, a fire-themed attendance day-streak card, and a clock-themed upcoming workout session timestamp tracker.
- Embed a wide promotional system slider item component right in the middle section of the viewport configured to cycle active gym marketing media announcements, business advertisements and call-to-action buttons.
- Create a dynamic filter pill container list spanning categories like strength, cardio, and weight lifts, directly driving a horizontally scrollable media display gallery showcasing interactive instructional exercise programming video tiles below it.
- Assemble a grid of quick-action component touch targets near the base of the page layout, mapping separate button blocks for loading the generated workout engine, pulling up new routines, spawning the geofence check-in scanner QR-code utility, and triggering the system training timer tool.

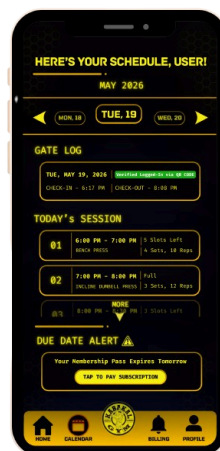
Wireframe Reference



Step 9: Calendar Page

- Set up the top layout view by placing a yellow header title over a short horizontal dividing line, and create a horizontal swipeable date strip element highlighting the current date in a thick gold-bordered capsule button flanked by directional navigation arrows pointing at the previous and following days
- Build the verified data-logging panel titled "GATE LOG" containing a rounded gold card component that tracks user attendance, displaying the current date, a prominent green verification badge indicator from what type/how the user logged-in (e.g., via QR-CODE), and precise numerical text blocks showing gym-user's check-in and check-out timestamps.
- Construct the training routine feed segment labeled "TODAY'S SESSION" using a vertical stack of rounded target class item cards, each featuring an index item sequence bubble on the left, time slots, specific exercise descriptions, sets and reps counts, and live capacity labels indicating if slots remain or if the time block is full.
- Incorporate a dynamic view expansion utility at the bottom of the session logs layout by rendering a fading text mask container labeled "MORE" alongside an interactive down-facing chevron arrow to toggle hidden/history calendar items.
- Assemble the bottom screen architecture by placing a bright high-visibility warning box titled "DUE DATE ALERT" that features an actionable yellow click pill labeled "TAP TO PAY SUBSCRIPTION", anchored neatly directly above the sticky gold-gradient system footer navbar.

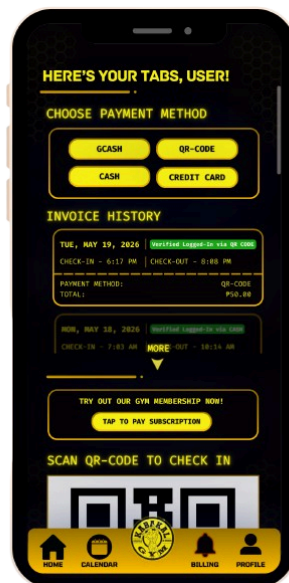
Wireframe Reference



Step 10:

- Set up the full-screen canvas layout with a dark matte black background like the previous pages and put subtle corner honeycomb textures, attaching the fixed gold-gradient bottom navigation bar with home, calendar, billing, and profile tabs.
- Create the header section by placing the yellow text like, "HERE'S YOUR TABS, USER!" above another title "CHOOSE PAYMENT METHOD" for the grid container that houses four solid yellow pill buttons for GCash, QR-Code, Cash, and Credit Card integrations
- Construct the scrollable "INVOICE HISTORY" module listing detailed receipt cards that track transactional data parameters, including the log date, a green verification check-in badge, raw check-in/out timestamps, selected payment method type, and the computed total balance due (₱50.00).
- Build a middle promotional banner box labeled "TRY OUT OUR GYM MEMBERSHIP NOW!" containing a primary, solid yellow call-to-action button template labeled "TAP TO PAY SUBSCRIPTION".
- Assemble the bottom interface block titled "SCAN QR-CODE TO CHECK IN/PAY" that displays a prominent QR code asset utilized for system geofence physical gate logs.

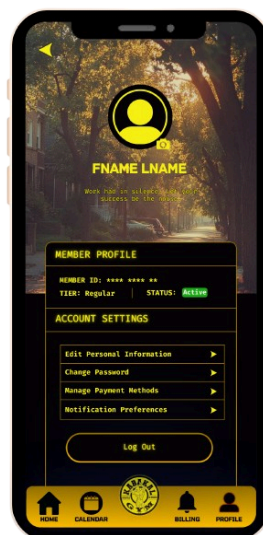
Wireframe Reference



Step 11:

- Create the background canvas by blending a frame cover image (which can be edited) with a solid matte black base section, attaching a yellow top-left back arrow icon and the fixed gold-gradient bottom navigation bar.
- Build the upper profile header component featuring a large yellow user avatar icon detailed with a small camera action badge for the user to edit their profile picture, stacked above the bold name heading "User's FirstNAME LastName" and a yellow motivational subtext quote (also can be edited depending on user's preference)
- Construct the "MEMBER PROFILE" information card displaying key account parameters, including a masked yellow text for Member's number ID, a Tier indicator, and a green rounded text for user's current status and/or membership badge.
- Create the "ACCOUNT SETTINGS" menu module structured as a vertical list of options containing clickable row items for "Edit Personal Information", "Change Password", "Manage Payment Methods", and "Notification Preferences", each styled with a right-facing yellow arrow chevron.
- Assemble the bottom action target container by positioning a pill-shaped, gold-bordered primary button reading "Log Out" directly beneath the settings list block

Wireframe Reference



[DESKTOP LAYOUT]

Week 3: Admin Interface

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Step –: Make sure the layout of every single page looks correct on small phones and larger computer screens

Step –: Add quick animation effects when clicking buttons so the user flow feels smooth

Step –: Run final build checks and package the finished code files for release